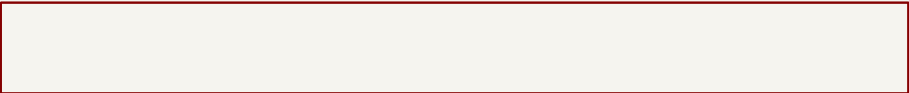


ALFRED / A1-4RD

5 V USB-C PD | removable Pi Zero 2 W | full-duplex I2S | one jaw servo

USB-C: fixed 5 V / 5 A contract



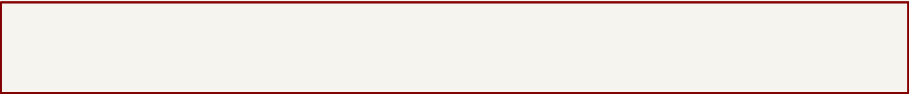
File: block1.kicad_sch

Protected rails and servo undervoltage override



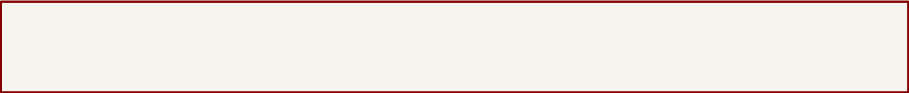
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Raspberry Pi physical header and shared I2S



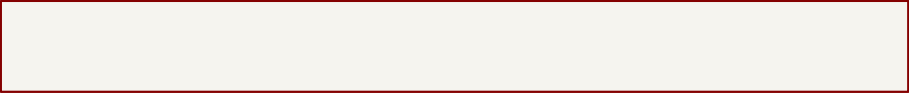
File: block3.kicad_sch

Digital microphone and bridge speaker amplifier



File: block4.kicad_sch

Jaw interface, energy dump, button and status



File: block5.kicad_sch

Hardware fault latch and revised voltage supervisor. Unbuilt prototype.
A1-4RD / Alfred

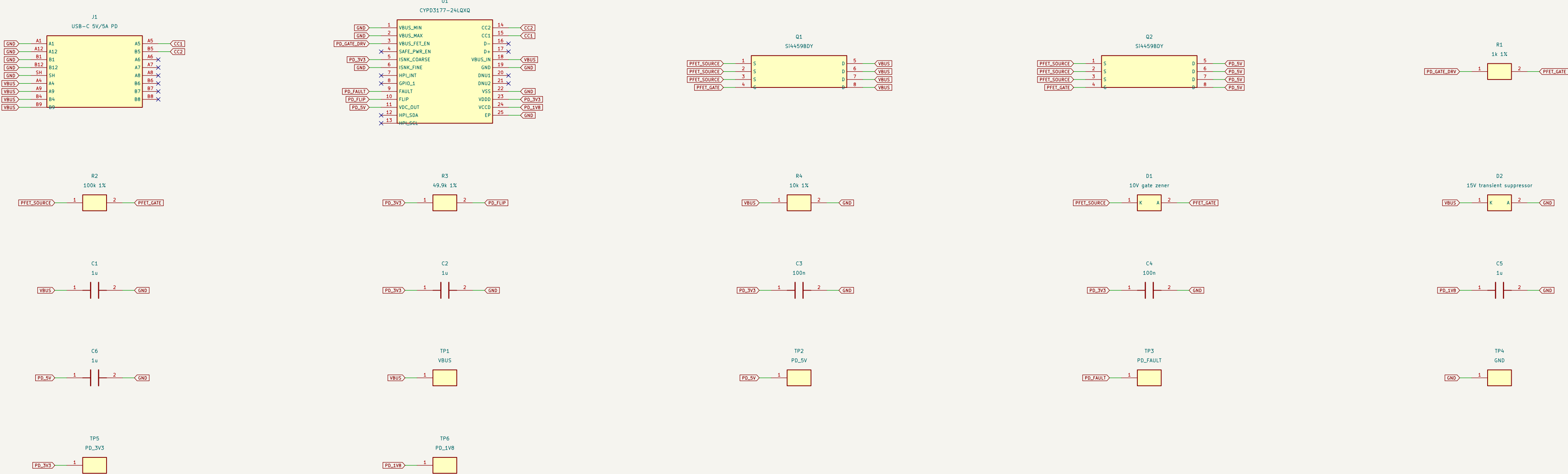
Sheet: /
File: Alfred.kicad_sch

Title: Alfred – one-servo thin client

Size: A4	Date: 2026-09-12	Rev: D – prototype
KiCad E.D.A. 10.0.6		Id: 1/6

USB-C: fixed 5 V / 5 A contract

Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Hardware fault latch and revised voltage supervisor. Unbuilt prototype.

A1-4RD / Alfred

Sheet: /USB-C: fixed 5 V / 5 A contract/

File: block1.kicad_sch

Title: USB-C: fixed 5 V / 5 A contract

Size: A2 Date: 2026-09-12

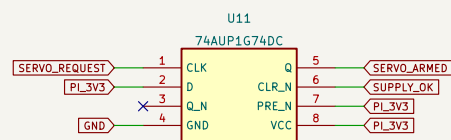
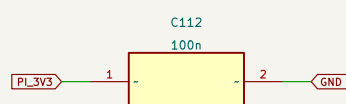
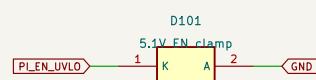
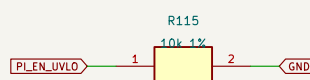
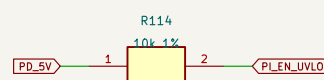
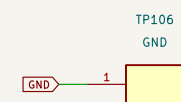
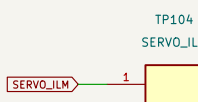
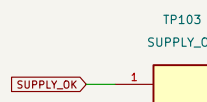
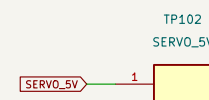
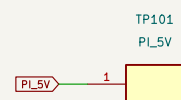
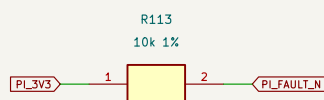
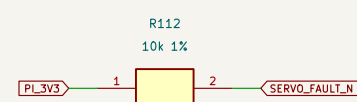
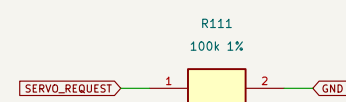
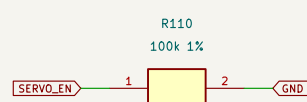
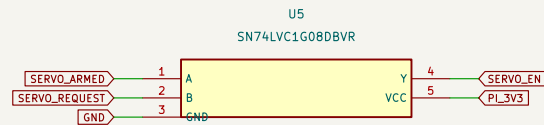
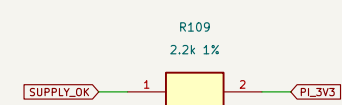
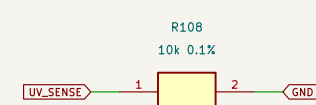
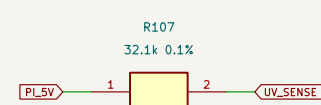
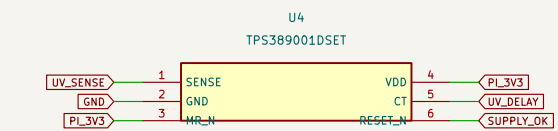
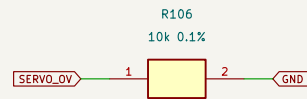
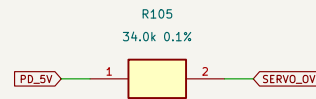
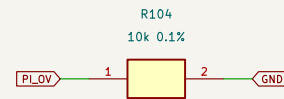
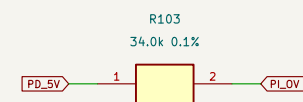
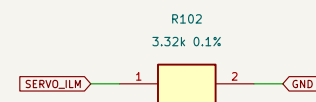
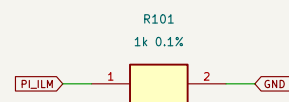
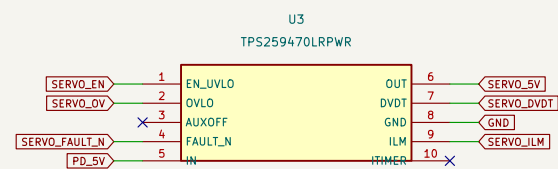
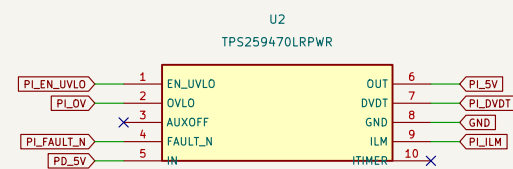
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Rev: D - prototype

Id: 2/6

Protected rails and servo undervoltage override

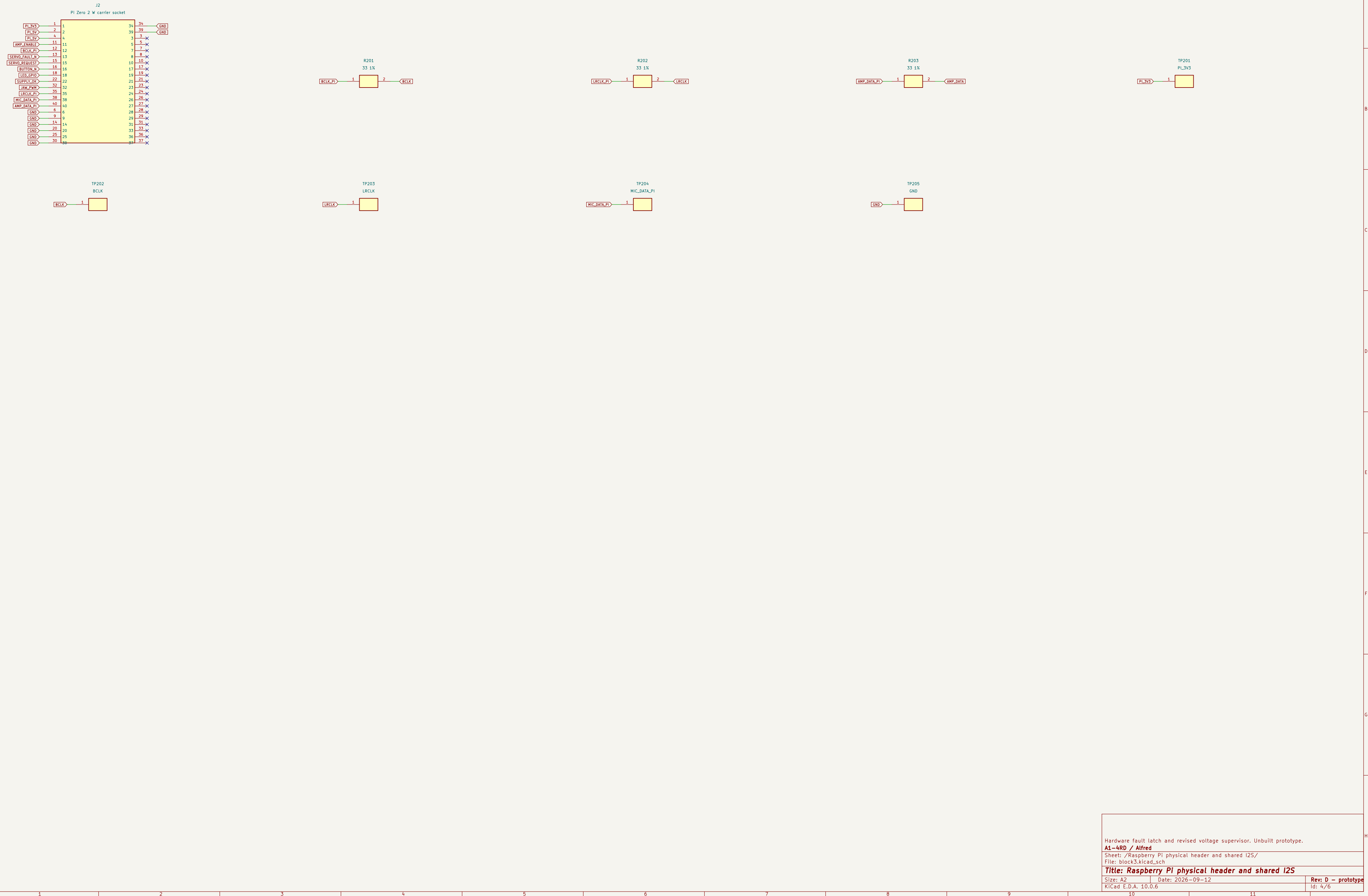
Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Hardware fault latch and revised voltage supervisor. Unbuilt prototype. A1-4RD / Alfred		
Sheet: /Protected rails and servo undervoltage override/ File: block2.kicad_sch		
Title: Protected rails and servo undervoltage override		
Size: A2	Date: 2026-09-12	Rev: D - prototype
KiCad E.D.A. 10.0.6		Id: 3/6

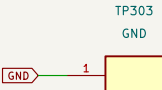
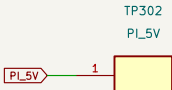
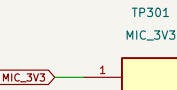
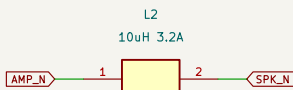
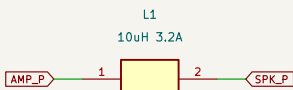
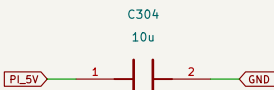
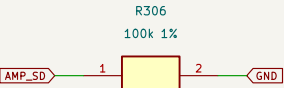
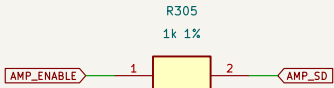
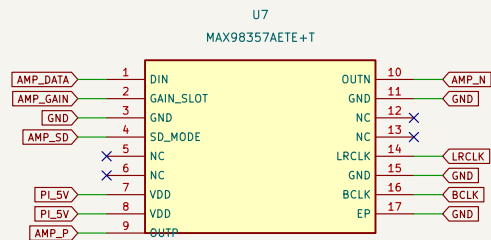
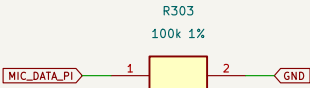
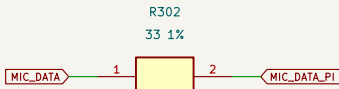
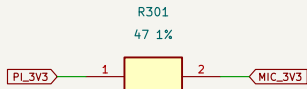
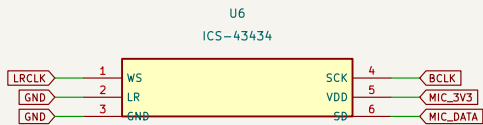
Raspberry Pi physical header and shared I2S

Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Digital microphone and bridge speaker amplifier

Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Hardware fault latch and revised voltage supervisor. Unbuilt prototype.

A1-4RD / Alfred

Sheet: /Digital microphone and bridge speaker amplifier/

File: block4.kicad_sch

Title: Digital microphone and bridge speaker amplifier

Size: A2

Date: 2026-09-12

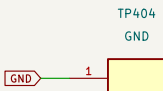
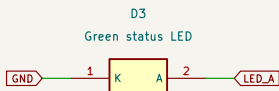
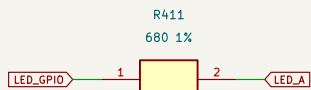
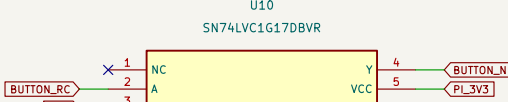
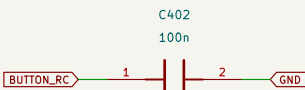
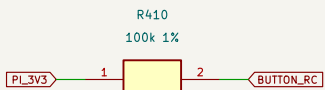
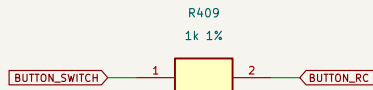
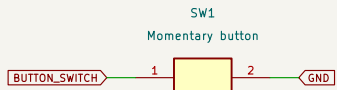
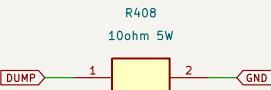
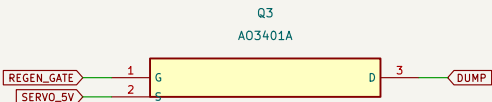
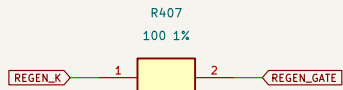
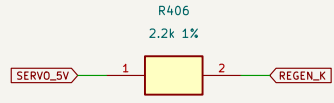
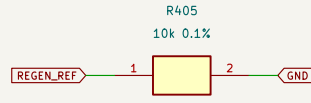
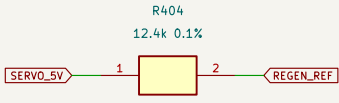
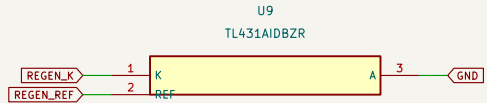
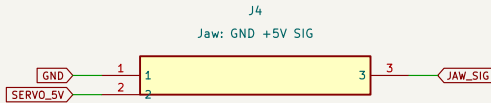
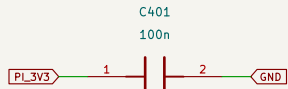
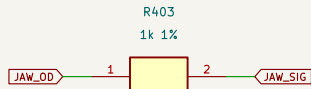
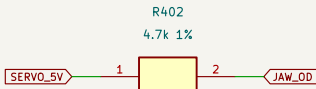
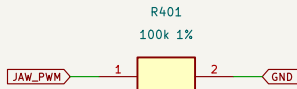
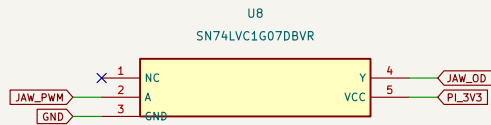
Rev: D - prototype

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Id: 5/6

Jaw interface, energy dump, button and status

Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Hardware fault latch and revised voltage supervisor. Unbuilt prototype.

A1-4RD / Alfred

Sheet: /Jaw interface, energy dump, button and status/

File: block5.kicad_sch

Title: Jaw interface, energy dump, button and status

Size: A2

Date: 2026-09-12

Rev: D - prototype

KiCad E.D.A. 10.0.6

Id: 6/6